06 — SRS_FEATURE_06_ADMIN_OPS_OBSERVABILITY_PILOT_RUNBOOK_TESTDATA_v1

> - Mục tiêu: vận hành pilot an toàn: admin tra cứu, flags/whitelist runtime, monitoring metrics/alerts, runbook xử lý sự cố, test data chuẩn cho staging/prod pilot.
> - In-scope: admin read-only, flags/whitelist controls, dashboard + alerts, pilot playbook.
> - Out-of-scope: admin override order state, BI nâng cao.
> - Rủi ro: không có visibility khi lỗi webhook/refund → giải pháp: metrics + alert + audit timeline.

1. INTRODUCTION

1.1 Purpose

- Đặc tả yêu cầu vận hành và quan sát hệ thống ordering.

1.2 Scope

- Ops/Admin tooling + monitoring + runbook + test data/pilot.

1.3 Definitions

- Kill-switch: PAYMENT_ENABLED OFF
- Whitelist mode: store-only hoặc store+user
- Evidence log: log/audit để điều tra

1.4 References

- Feature 04 (logging fields)
- Feature 01 (flags/whitelist)

2. OVERALL DESCRIPTION

2.1 Product Perspective

- Feature này giúp “release không mù”.

2.2 Product Functions

- Admin list/detail
- Flags/whitelist runtime updates (audit)
- Dashboard metrics
- Alerts
- Pilot rollout checklist
- Test data seed plan

2.3 User classes

- Ops
- QA
- PO/Sếp (read summary)

2.4 Operating environment

- Web admin
- Monitoring stack (Grafana/Datadog/CloudWatch…) tùy hiện trạng

2.5 Constraints

- Không được can thiệp sai vào giao dịch (read-only order/payment/refund)
- Mọi change flags/whitelist phải audit

2.6 Dependencies

- Logging/metrics pipeline tồn tại
- Admin auth/role available

3. SPECIFIC REQUIREMENTS

3.1 External Interface Requirements

- Admin UI: Orders list, order detail, flags/whitelist page, metrics overview (link)
- API:
  - GET /admin/orders
  - GET /admin/orders/{id}

- (Optional) POST /admin/flags, POST /admin/whitelist

3.2 Functional Requirements

FR-601 Admin Order List (Read-only)

- Filters: date range, store_id, status, payment status, refund status, keyword q
- Output columns: order_id/code, store_id, status, total_amount, created_at, paid_at, refund_status
- Pagination required

FR-602 Admin Order Detail (Read-only)

- Must show:
  - snapshot items, totals
  - payment attempts (status, intent_id, expires_at)
  - refund record (status, refund_id, amount, reason)
  - audit timeline (ordered)
- Output: 1 trang đủ để triage incident

FR-603 Flags Runtime Control

- Flags: ORDERING_ENABLED, PAYMENT_ENABLED
- Rules:
  - The update must take effect immediately
  - Every update writes audit: flag_changed(who, when, old→new)

FR-604 Whitelist Runtime Control

- Manage:
  - whitelist_store_ids
  - whitelist_user_ids (optional)
  - whitelist_mode
- Audit required: whitelist_changed

FR-605 Kill-switch Playbook (Operational)

- Trigger conditions: pending spike, webhook error spike, refund spike
- Action: set PAYMENT_ENABLED=OFF

- Output:
  - create payment blocked instantly
  - Ongoing refunds are still processed
- Evidence: log entry kill_switch_activated

FR-606 Monitoring Dashboard (Minimum metrics)

- Metrics:
  - payment_succeeded/failed/pending/expired
  - webhook_invalid_signature
  - webhook_duplicate
  - refund_requested/succeeded/failed
  - merchant_time_to_accept p50/p95
  - sla_auto_cancel_count
- Output: 1 dashboard link

FR-607 Alerts (Minimum)

- pending spike
- refund spike
- webhook error spike
- refund_failed > 0 sustained
- Output: alert routed to ops channel (email/telegram/lark…)

FR-608 Test Data & Pilot Seed Plan

- Required seed:
  - 1–3 pilot stores verified/active
  - each store: 2 categories, 3+ SKUs with valid price, 1 SKU unavailable
  - internal users, merchant account mapping store
- Output: staging E2E pass; prod pilot ready

FR-609 Prod Pilot Runbook (Option A)

- Steps:
  a. Deploy backend + migrate (flags OFF)
  b. Verify webhook signature prod

c. Verify jobs 10’ & 15’ with test order evidence
d. Add whitelist
e. Enable ORDERING then PAYMENT
f. Monitor 4–6h
g. Drill kill-switch

- Output: release controlled and auditable

3.3 Non-Functional Requirements

- NFR-601 Auditability: every control action traceable (who/when/what)
- NFR-602 Security: role-based access (ops only)
- NFR-603 Availability: dashboard accessible during the incident
- NFR-604 Performance: admin list p95 < 1s (pilot)

Dưới đây là **phần document bổ sung (appendix) để chèn trực tiếp vào SRS hiện tại**, mục tiêu là **để Design có thể làm Figma ngay**.

Bạn có thể đặt section này **ngay sau phần 3.2 Functional Requirements** hoặc tạo **một chương mới**:

Code block

```
4. ADMIN UI SPECIFICATION (FOR DESIGN)
```

Doc này **không thay đổi SRS hiện tại**, chỉ **bổ sung UI/UX spec cho Web Admin**.

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4. ADMIN UI SPECIFICATION

4.1 Purpose

Section này mô tả chi tiết **các màn hình Admin Web cần thiết để hỗ trợ vận hành pilot của hệ thống ordering**.

Mục tiêu:

- Design có thể tạo Figma dựa trên screen spec
- Dev hiểu layout và interaction
- QA hiểu UI state để viết test case

Các màn hình cần thiết:

1. Orders List
2. Order Detail
3. Flags Control
4. Whitelist Control
5. Monitoring Entry
6. Pilot Runbook

4.2 Admin Navigation

Admin navigation:

Code block

```
Admin
 ├ Orders
 ├ Runtime Controls
 │   ├ System Flags
 │   └ Whitelist
 ├ Monitoring
 └ Pilot Runbook
```

4.3 Orders List Screen

Purpose

Cho phép Ops tra cứu order nhanh khi có sự cố.

Layout Structure

Code block

```
--------------------------------------------------
Admin Header
--------------------------------------------------

Orders

```

```
--------------------------------------------------
Filters Bar
--------------------------------------------------
Date range | Store | Status | Payment | Refund | Search

--------------------------------------------------
Orders Table
--------------------------------------------------

order_id | store_id | status | total | payment | refund | created

--------------------------------------------------
Pagination
--------------------------------------------------
```

Filters

Filter	Type
Date range	date picker
Store	dropdown
Order status	dropdown
Payment status	dropdown
Refund status	dropdown
Keyword	text input

Table Columns

Column	Description
Order ID	clickable link
Store ID	store identifier
Status	order status
Total Amount	total bill
Payment Status	payment state
Refund Status	refund state
Created Time	timestamp

Interaction

Click **Order ID**

→ open **Order Detail page**

States

Loading

Code block

```
Loading orders...
```

Empty

Code block

```
No orders found
```

Error

Code block

```
Failed to load orders
Retry
```

4.4 Order Detail Screen

Purpose

Trang này phải đủ thông tin để Ops **debug incident trong 1 trang duy nhất**.

Layout Structure

Code block

```
--------------------------------------------------
Order Summary
--------------------------------------------------

Order ID
Store
User
Status
Created Time
Total Amount

--------------------------------------------------
Items Snapshot
--------------------------------------------------

item | price | qty | subtotal

--------------------------------------------------
Payment Attempts
--------------------------------------------------

attempt | status | intent_id | expires_at

--------------------------------------------------
Refund Information
--------------------------------------------------

refund_status | refund_id | amount | reason

--------------------------------------------------
Audit Timeline
--------------------------------------------------

timestamp | event | metadata
```

Section 1 – Order Summary

Field
Order ID
Store
User
Status
Created Time
Total Amount

Section 2 – Items Snapshot

Column
Item name
Unit price
Quantity
Subtotal

Section 3 – Payment Attempts

Column
Attempt ID
Status
Intent ID
Expires At

Section 4 – Refund Information

Field
Refund status
Refund ID
Refund amount
Reason

Section 5 – Audit Timeline

Timeline hiển thị sự kiện theo thứ tự thời gian.

Ví dụ:

Code block

```
12:00 order_created
12:01 payment_initiated
12:02 payment_succeeded
12:10 merchant_accepted
```

4.5 System Flags Control Screen

Purpose

Cho phép Ops bật/tắt ordering hoặc payment **runtime**.

Layout

Code block

```
--------------------------------------------------
System Flags
--------------------------------------------------

ORDERING_ENABLED      [Toggle ON/OFF]

PAYMENT_ENABLED       [Toggle ON/OFF]

--------------------------------------------------
Audit History
--------------------------------------------------

time | user | flag | old_value | new_value
```

Interaction

Khi toggle flag:

Hiển thị confirm modal:

```
Confirm change?

Disable PAYMENT_ENABLED?
```

Audit

Sau khi thay đổi:

```
Changed by: admin@email
Time: timestamp
```

4.6 Whitelist Control Screen

Purpose

Cho phép rollout pilot theo store hoặc user.

Layout

```
--------------------------------------------------
Whitelist Mode
--------------------------------------------------

Disabled
Store Only
Store + User

--------------------------------------------------
Stores
--------------------------------------------------
```

```

store_id list

[ Add Store ]

--------------------------------------------------
Users
--------------------------------------------------

user_id list

[ Add User ]
```

Interaction

Add Store

Code block

```
Input store_id
Save
```

Add User

Code block

```
Input user_id
Save
```

Audit Event

Mọi thay đổi whitelist phải log:

Code block

```
whitelist_changed
```

4.7 Monitoring Entry Screen

Admin Web **không cần build dashboard phức tạp**.

Chỉ cần entry point.

Layout

Code block

```
--------------------------------------------------
Monitoring
--------------------------------------------------

Payment Metrics
[Open Dashboard]

Webhook Metrics
[Open Dashboard]

Refund Metrics
[Open Dashboard]

Alerts
[View Alerts]

--------------------------------------------------
System Status
--------------------------------------------------

Ordering Enabled
Payment Enabled
```

4.8 Pilot Runbook Screen

Trang này hiển thị checklist rollout.

Layout

```
--------------------------------------------------
Pilot Release Checklist
--------------------------------------------------

[ ] Backend deployed
[ ] Flags OFF verified
[ ] Webhook signature verified
[ ] Job scheduler verified
[ ] Whitelist configured
[ ] ORDERING_ENABLED enabled
[ ] PAYMENT_ENABLED enabled
[ ] Monitoring verified
```

4.9 Design Principles

Simplicity

Admin UI cần đơn giản vì thường được dùng khi **incident xảy ra**.

Fast Navigation

Order detail phải mở được **trong 1 click từ order list**.

Evidence First

Order detail phải hiển thị:

- payment attempts
- refund records
- audit timeline

Safe Control

Flags control phải có:

- confirm modal
- audit log

4.10 Screens Required for Figma

Design cần tạo tối thiểu **6 screens**:

1. Orders List
2. Order Detail
3. System Flags Control
4. Whitelist Control
5. Monitoring Entry
6. Pilot Runbook

4.11 Design Deliverables

Design team cần cung cấp:

- Desktop layout (Admin Web)
- Table design
- Timeline component
- Toggle control states
- Modal confirm design

Kết luận

Section này bổ sung:

- **screen layout**
- **UI components**
- **interaction**
- **states**

để **Design có thể tạo Figma trực tiếp từ SRS**.

Nếu bạn muốn, tôi có thể viết thêm **1 phần nữa cực quan trọng cho design + dev + QA**:

Admin Order Detail Layout chuẩn giống Stripe / Shopify

Layout này giúp **Ops debug incident nhanh hơn 3–5 lần**, và thực tế **90% hệ thống order production đều dùng pattern này.**